

Replay-Aware ATSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: atsp_failure_replay_compression.
- Best TSPLIB ATSP holdout gap: atsp_failure_replay_compression.
- Best synthetic holdout gap: atsp_random_replay.

Run Metadata

- run_name: run_20260513_184033_r
- started_at_local: 2026-05-13 18:40:33
- finished_at_local: 2026-05-13 18:49:09
- duration_hhmm: 00:09
- duration_seconds: 516.817
- seed_offset: 17000
- replicate_label: r
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB ATSP Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
atsp_no_replay	none	score_only	False	0.212881	0.02698	0.11993	0.699104	0.78	0.022781
atsp_random_replay	random	score_only	False	0.193105	0.008454	0.100779	0.0	0.58	0.0
atsp_failure_replay	failure	score_only	False	0.202334	0.034191	0.118263	0.0	0.78	0.0
atsp_failure_replay_compression	failure	novelty_gate	True	0.150416	0.041579	0.095997	0.0	0.78	0.0

Condition Notes

atsp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.212881, synthetic 0.02698, combined 0.11993.

- Accepted-epoch count 2, mean accepted code novelty 0.699104, and final complexity 0.78.
- Adaptation efficiency 0.022781 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.212881 across 4 instances; family means: ft=0.278349, ftv=0.345939, p=0.003203, ry=0.224033.
- Panel synthetic_holdout mean gap 0.02698 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.041205, wind_clusters=0.066716.
- Worst recent training cases: [{"best_known_cost": 1286, "cost": 1607, "family": "ftv", "name": "ftv33", "optimality_gap": 0.249611}, {"best_known_cost": 1473, "cost": 1821, "family": "ftv", "name": "ftv35", "optimality_gap": 0.236253}, {"best_known_cost": 1530, "cost": 1842, "family": "ftv", "name": "ftv38", "optimality_gap": 0.203922}]

atsp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.193105, synthetic 0.008454, combined 0.100779.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.58.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.193105 across 4 instances; family means: ft=0.323244, ftv=0.33478, p=0.004982, ry=0.109416.
- Panel synthetic_holdout mean gap 0.008454 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1939, "family": "ftv", "name": "ftv38", "optimality_gap": 0.26732}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1705, "family": "ftv", "name": "ftv35", "optimality_gap": 0.157502}]

atsp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.202334, synthetic 0.034191, combined 0.118263.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.78.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.202334 across 4 instances; family means: ft=0.425923, ftv=0.270304, p=0.000712, ry=0.112398.
- Panel synthetic_holdout mean gap 0.034191 across 4 instances; family means: clockwise_ring=0.085613, corridor_drift=0.0, hub_spokes=0.051151, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1286, "cost": 1640, "family": "ftv", "name": "ftv33", "optimality_gap": 0.275272}, {"best_known_cost": 1530, "cost": 1863, "family": "ftv", "name": "ftv38", "optimality_gap": 0.217647}, {"best_known_cost": 1473, "cost": 1758, "family": "ftv", "name": "ftv35", "optimality_gap": 0.193483}]

atsp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB ATSP 0.150416, synthetic 0.041579, combined 0.095997.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.78.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.

- Panel heldout_tsplib mean gap 0.150416 across 4 instances; family means: ft=0.203765, ftv=0.295102, p=0.001423, ry=0.101373.
- Panel synthetic_holdout mean gap 0.041579 across 4 instances; family means: clockwise_ring=0.085613, corridor_drift=0.0, hub_spokes=0.080705, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1473, "cost": 1943, "family": "ftv", "name": "ftv35", "optimality_gap": 0.319077}, {"best_known_cost": 1286, "cost": 1649, "family": "ftv", "name": "ftv33", "optimality_gap": 0.282271}, {"best_known_cost": 1530, "cost": 1857, "family": "ftv", "name": "ftv38", "optimality_gap": 0.213725}]

Judge Appendix

Summary of Results by Replay Condition

Condition	Replay Mode	Compression Pressure	TSPLIB Gap (mean)	Synthetic Gap (mean)	Transfer Gap (mean)	Code Novelty (mean)	Complexity (mean)
atsp_no_replay	None	No	0.2129	0.0270	0.1199	0.6991	0.78
atsp_random_replay	Random	No	0.1931	**0.0085**	0.1008	0	0.58
atsp_failure_replay	Failure	No	0.2023	0.0342	0.1183	0	0.78
atsp_failure_replay_compression	Failure + Compression	Yes	**0.1504**	0.0416	**0.0960**	0	0.78

Interpretation

- **Main Transfer Metrics** (held-out TSPLIB and synthetic gaps):
- The **lowest mean TSPLIB gap (0.1504) and transfer gap (0.0960)** occur under the **atsp_failure_replay_compression** condition.
- Synthetic holdout gap is best under **atsp_random_replay** (0.0085), but transfer gaps are slightly worse than failure replay with compression.
- **No replay** has higher TSPLIB and transfer gaps despite high code novelty, indicating that code novelty does not translate to better transfer.
- Pure **failure replay without compression** shows modest improvements over no replay but worse than failure replay with compression.
- **Code Novelty vs Transfer:**
- Code novelty is **high only in no replay** (around 0.7), but this coincides with **worst TSPLIB and transfer gaps**.
- All replay conditions with replay (random, failure, failure+compression) have **zero code novelty**, indicating no new synthetic structural variation introduced.
- Despite zero code novelty, failure replay with compression outperforms others in transfer metrics, suggesting that improved transfer is achieved through replay mechanisms rather than new code innovation.
- **Effect of Replay Modes:**
- **No replay:** High code novelty but poor transfer and TSPLIB performance.
- **Random replay:** Best synthetic holdout gap, moderate transfer and TSPLIB gaps, no code novelty.

- **Failure replay:** Slightly worse than random replay for synthetic and TSPLIB gaps, no code novelty.
- **Failure replay + compression:** Best transfer and TSPLIB performance, highest complexity (same as other failure replay), no code novelty.
- **Complexity and Archive Sizes:**
- All conditions maintain similar archive sizes (11 total), with similar distribution among adversarial, failure, worst cases.
- Complexity is highest (0.78) in no replay and failure replay variants, lowest (0.58) in random replay.
- Compression pressure is only applied in failure replay with compression.

Conditional Ranking by Transfer (TSPLIB held-out gap prioritized)

1. **atsp_failure_replay_compression:** Best transfer and held-out TSPLIB gap (0.0960, 0.1504), no code novelty.
2. **atsp_random_replay:** Second best transfer (0.1008), best synthetic holdout gap (0.0085), no code novelty.
3. **atsp_failure_replay:** Moderate transfer (0.1183), higher TSPLIB gap (0.2023), no code novelty.
4. **atsp_no_replay:** Worst transfer (0.1199) and TSPLIB gap (0.2129), highest code novelty (0.6991).

Conclusions

- **Replay improves transfer metrics compared to no replay**, with **failure replay combined with compression yielding the best held-out TSPLIB and transfer gaps**.
- This improvement occurs **despite zero code novelty**, meaning replay mechanisms and compression, rather than algorithmic innovation, drive generalization gains.
- **Random replay achieves the best synthetic holdout gap but slightly worse transfer**, indicating synthetic performance does not always translate to transfer.
- **No replay condition's high code novelty does not correspond to better transfer or TSPLIB performance**, suggesting code novelty here reflects non-effective exploration.
- Distinguishing replay modes is critical: **failure replay with compression is superior to failure replay alone** and both surpass no replay.
- Prioritize **atsp_failure_replay_compression** condition for best evidence of transfer improvement.

Key Metrics for Best Transfer Condition (atsp_failure_replay_compression)

- TSPLIB gap: 0.1504 (best)
- Transfer gap: 0.0960 (best)
- Synthetic gap: 0.0416 (worst among replay, but acceptable)
- Code novelty: 0 (lowest)
- Complexity: 0.78 (high)
- Replay mode: failure replay + compression pressure enabled

This conservative interpretation focuses on transfer gaps and shows that replay (especially failure replay with compression) enhances generalization despite no accompanying code novelty increase.